

Anya Lee

408-809-6652 | anyalee6810@gmail.com | [linkedin.com/in/anya-lee/](https://www.linkedin.com/in/anya-lee/) | github.com/mathstud

EDUCATION

University of Colorado, Boulder

Boulder, CO

Master of Science in Applied Mathematics, Option in Statistics and Data Science, GPA 4.0

May 2026

Oregon State University

Corvallis, OR

Bachelor of Science in Mathematics, Option in Statistics, GPA 3.90

Jun. 2024

EXPERIENCE

Course Assistant

Jan. 2026 – May 2026

University of Colorado, Boulder

Boulder, CO

- Attended lecture for STAT 5000: Statistical Methods & Applications I through entire semester
- Created and posted R and RStudio coding tutorials
- Held three office hours each week for student questions and help
- Created homework assignments and rubrics for grading
- Proctored midterm and final exam; re-scheduled those with time conflicts
- Conducted grade entry and provided constructive commentary
- Met with instructor about grading preferences, exams, and other assessments

Graduate Research Assistant

Jun. 2025 – Jul. 2025

University of Colorado, Boulder

Boulder, CO

- Built four reusable Python notebooks that load, pre-process, and analyze Optimum Interpolation Sea Surface Temperature (OISST) yearly time series data from 15,979 NetCDF files to save time for future analyses by supervisor
- Applied fixed- and variable-scale spatial smoothing algorithms from GitHub with Python to improve data quality and enhance climate signal detection
- Analyzed Sea Surface Temperature data to identify Marine Heat Wave (MHW) events through statistical metric computation, and time-series and geospatial plots and visualization
- Leveraged open-source Python libraries (gcm-filters and marineHeatWaves) to develop reproducible climate data analysis workflows

Software Engineer Intern

Jun. 2023 – Sep. 2023

Robust Intelligence, Inc.

San Francisco, CA

- Created and tested a reusable Jupyter Notebook that assists engineers in their demonstrations of Generative AI stress testing for efficient customer understanding
- Executed the notebook using API tokens and validated output to ensure successful product demonstration
- Wrote a python script to generate “Related Topics” links to the bottom of company web documentation to improve user navigation
- Identified and fixed bugs in the documentation code to enhance user readability
- Presented internship learnings and accomplishments to the company

PROJECTS

Evaluating Pretrained Architectures for Agricultural Pest Classification

Apr. 2026 – May 2026

- Conducted comparative analysis between two experiments of zero shot inference and fine-tuning with a pretrained convolutional neural network (CNN) and vision transformer (from PyTorch and Hugging Face) in Python
- Preprocessed image dataset of 15 classes with 100 images per class
- Wrote a 13-page paper with experiment results and presented an academic poster

Exploring Unsupervised Methods of Identifying Cybersecurity Events

Aug. 2025 – Dec. 2025

- Developed a scalable unsupervised pipeline (5 pathways) for identifying cybersecurity events from 13,079 online news articles without manual data annotation to quantify threat risks
- Implemented Hugging Face Transformer models to obtain word embeddings for NLP classification tasks
- Performed data cleaning, clustering on model embeddings, visualization of clustering results, and cluster analysis that resulted in meaningful groupings to provide insight on preventing future cyberattacks

- Submitted 10+ batch jobs to run code scripts on dedicated compute nodes through Alpine, the University of Colorado Boulder Research Computing's High Performance Computing cluster
- Collaborated in a team of four that wrote a 13-page paper and verbally communicated findings with a 23-slide presentation to the Applied Mathematics Department, Capstone course, and Sandia National Labs project sponsor

Multi-Strain Modeling the Impact of Cross-Immunity on Infection Dynamics Nov. 2025 – Dec. 2025

- Developed an extension of the SIR compartmental model to simulate a two-strain system with a focus on the effect of a vaccine providing cross immunity to understand population infection dynamics
- Programmed a Python script to simulate the system of differential equations to observe the effects of vaccination
- Wrote a 7-page paper with simulation results and presented to the class

Prevalence of Diabetes by Country: A Statistical Modeling Approach Mar. 2025 – May 2025

- Investigated the extent that region and income level of a country affect prevalence of diabetes when accounting for diabetes risk factors like lifestyle factors (i.e. BMI) and socioeconomic factors (i.e. GDP per capita)
- Performed data cleaning, merging, and imputation from several data sources
- Modeled and compared linear fixed effects model in R to analyze relevant factors
- Wrote paper and presented methodology and results

Baseball Statistical Analysis Nov. 2024 – Dec. 2024

- Analyzed MLB player performance and salary data for batters, pitchers, and fielders using statistical models to identify key predictors of strikeouts, ERA, and compensation across positions
- Investigated data relationships and anomalies through exploratory analysis and regression modeling, translating results into clear insights for a technical and non-technical audience
- Presented analytical findings on performance metrics and salary predictors with a 30-slide presentation as a final project for STAT 5010

TECHNICAL SKILLS

Languages: Python, R, SQL, Markdown, Scala, Apache Spark

Developer Tools: Git, GitHub, Tableau, PowerBI, JupyterLab, Google Workspace (e.g. Google Sheets), Microsoft Office (e.g. Excel), VS Code, Slack, Notion, Zoom

Libraries: CuPy, pandas, NumPy, Matplotlib, keras, PyTorch, TensorFlow, Scikit-learn, scipy, seaborn, torchvision

Courses: Methods of Data Analysis, Machine and Deep Learning, Statistical Learning and Modeling, Time Series, Big Data Tools